



Life beneath *the* Waves

Oceans cover nearly 70 percent of this planet, making them the largest of all habitats. There are plenty of creatures living beneath the waves, but life here presents a very different set of challenges to living on the land.

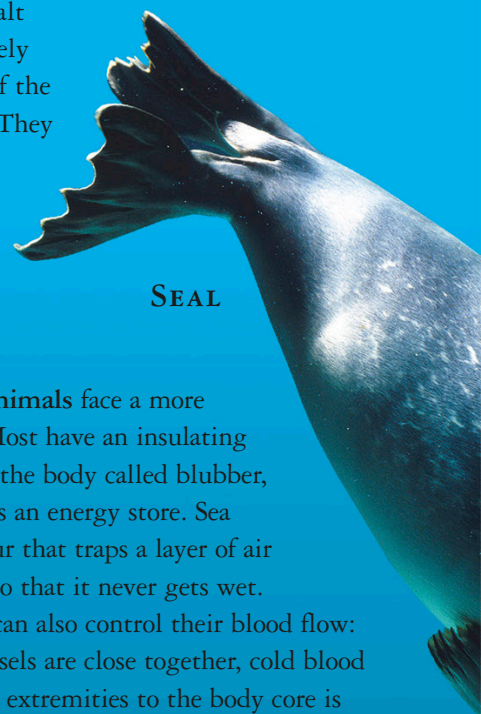
IT'S SALTY
DOWN HERE!

*Seawater contains
35 parts of dissolved salts
for every 1,000 parts of water.*



It's too salty

Even though the salt in seawater is heavily diluted, it would damage any land animal or plant that tried to drink it. However, the water inside the bodies of marine organisms is as salty as the water around them, which keeps them in balance. Fish drink seawater but they secrete excess salt through their gills. Mammals rarely drink seawater and obtain most of the water they need from their food. They get rid of the salt in their urine.

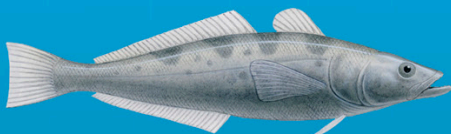


SEAL

Keeping warm

The surface layers of the ocean can be very warm, especially near the coast where the water is shallower. In the deep water and around the poles it is extremely cold. Most marine life is cold-blooded, matching the temperature of the water around it. Sometimes, however, that is too cold—there are fish living in polar waters that have a type of antifreeze in their blood to keep it flowing.

ANTARCTIC TOOTHFISH



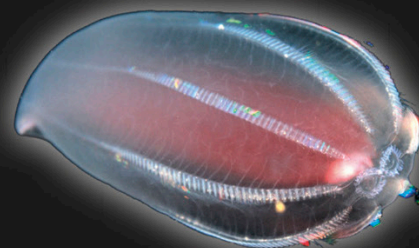
Warm-blooded animals face a more severe challenge. Most have an insulating layer of fat around the body called blubber, which also serves as an energy store. Sea otters have thick fur that traps a layer of air next to their skin so that it never gets wet. Marine mammals can also control their blood flow: since the blood vessels are close together, cold blood returning from the extremities to the body core is warmed by blood flowing back out to the limbs.

ANGLER FISH



Swimming in the dark

Sunlight cannot penetrate more than about a yard through water, so organisms that rely on photosynthesis, such as corals and seaweed, are found in shallow water. It gets much darker as you go deeper. It is hard to see down here, so animals rely on other senses—smell, echolocation, hearing, and changes in water pressure—to detect food and enemies. Some animals produce their own light—called bioluminescence—to attract prey or mates.



COMB JELLYFISH